

Homework #4

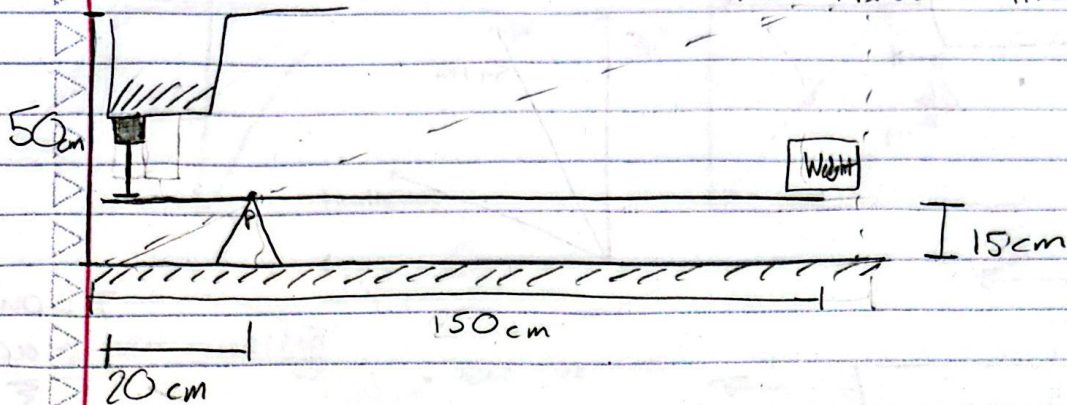
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Cole Finnan

Statics

Portfolio Problem: Actuator: IMA44 RN10

• Max Thrust = 11.03 kN



$$\sum M_p = 11.03(20) - W(1.2) = 0$$
$$2.206 = W(1.2)$$

$$W = 1.838 \text{ kN}$$

Mass

$$\theta = \tan^{-1}\left(\frac{15}{20}\right) = 36.87^\circ$$

$$\text{Max weight} = 1.838 \text{ kN}$$

$$\text{Max height above the top of the pivot} = 0.975 \text{ m}$$

$$\frac{15}{20} = \frac{h}{130}$$

$$h = 97.5 \text{ cm}$$